Ala Bouhaouel

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Github: https://github.com/AlaaBouhaouel

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Personal Summary:

Passionate high school graduate and ISEF 2024 award recipient with a strong interest in software engineering and artificial intelligence. Eager to join research-focused academic environments and contribute to impactful tech innovation.

Education:

- *High School of 2 Mars 1934, Tunisia:* Baccalaureate in Mathematics |**Juin, 2025**
  Ranked **1st** in graduating class with a score of **17.41/20**
  Relevant Modules: Mathematics, Computer Science, Physics

Awards and Achievements:

- Finalist and Special Award Winner, ISEF (Regeneron International Science and Engineering Fair) **| May, 2024**
- Mawhiba Full Scholarship Awardee – King Fahd University of Petroleum & Minerals (KFUPM) **|during ISEF as a special award**
- Two times Gold medalist and top 10 in I-Fest (National Science Fair in Tunisia) **|Mars, 2023 and Mars, 2024**
- Gold Medal – ICAN 2023 (International Invention Innovation Competition in Canada, organized by TISIAS, Toronto) **|Aug 2023**
- Best Young Inventor and Special Award – ICAN, TISIAS **|Aug 2024**
- Silver medal in Genius Olympiad USA 2023 (RIT scholarship) **|Jun 2023**
- Bronze medal in IOAI in Bulgarie **| Aug, 2024**
- Gold medal in AI Youth Challenge Hackathon in Tunisia **|Oct, 2023**
  (Selected among Top 5 finalists to represent Tunisia internationally in IOAI- Bulgarie 2024)

Projects and experiences:

❖ ***Founder & Lead Engineer – OLIVY (Prototype Stage):*** **|from 2023 to 2024**

Research paper will be attached (ISEF 2024 – OLIVY Project).
Designed and led development of OLIVY, an AI-powered platform integrating drone automation and machine learning to detect, map, and autonomously treat olive tree diseases. Engineered a precision agriculture system with 92% detection accuracy, real-time geospatial analytics and chatbot integration. Conducted field validation with farmers and experts to ensure usability and real-world application.

❖ ***Other experiences and extracurricular activities:***

- 2 STEM summer camps and 2 STEM winter camps (**AI – IOT- Web development (Django) -App development (Flatter)**) **| Aug, 2022 |Dec, 2023| Aug, 2024**
- Volunteer in vex competition **|Feb, 2023**
- Volunteer in I-Fest **|Mars, 2024**
- Membership in STEM club "ATAST" for 3 years. **|2022 to 2025**

- ATAST Tunisia Website developer (In Development):

  Designed and developed the future official website for the Tunisian Association for The Future Of Science And Technology (ATAST). Built using Django, HTML/CSS/Bootstrap. Reviewed and informally approved by the ATAST founder; publication pending. |Source code: github.com/AlaaBouhaouel/Atast_website

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Skills:

❖ **AI & Machine Learning Skills:**

Algorithms: Logistic Regression, Linear/Polynomial Regression, KNN, Random Forest
Deep Learning: Neural Networks, CNNs, Transfer Learning (Keras, TensorFlow)
Feature Engineering & Data Prep: Scaling, Encoding, Aggregation, Normalization
Computer Vision: Image classification, preprocessing, defect detection (OpenCV)
NLP: Tokenization, Text Preprocessing, Word Embeddings, Text Classification
Model Evaluation & Optimization: Confusion Matrix, Precision/Recall, Hyperparameter Tuning

❖ **Other Technical Skills:**
Programming languages (Python, C++)
Tools and frameworks (scikit-learn, Keras, TensorFlow, Arduino)
Web Development: Django (Python), HTML, CSS, Bootstrap
Backend: Django REST Framework, SQLite

Certifications And Courses:

- Certificate of Completion **– World SDGs with MIEEs** Program (Global Sustainability & Innovation)**, Korean International School in HCMC | Jun, 2023**
- Certificate of Achievement – **35h Artificial Intelligence & Robotics Training |Aug, 2023**
- Certificate of Participation & Volunteering – **VEX Robotics Competition**, Tunisia National Finals **|Feb, 2023**
- Certificate of Participation – **Genius Olympiad**, Science Category **|Feb, 2023** (Selected among Top 5 finalists to participate in the final round in New York)
- Certificate of Participation – **Z-Generation Innovation Forum 2.0: "Living on Mars" |Apr, 2024**
- Diploma, **Coding** Level 3 – Inventors Category (12h) **|Jul, 2023**
- Diploma, **Coding** Level 4 – Inventors Category (12h) **|Aug, 2022**
- Diploma, **Robotics Coding** Level 5 – Inventors Category (12h) **|Sep, 2022**
- Diploma, **3D Design** Level 1 – Inventors Category (12h) **|Jul, 2022**
- Diploma, Robotics Summer Training – **3D Design & Coding** Level 3 (36h) **|Aug, 2023**

Please find attached my CV along with a selection of supporting certificates, including ISEF, Mawhiba, and AI/robotics training. Additional certificates are available upon request.